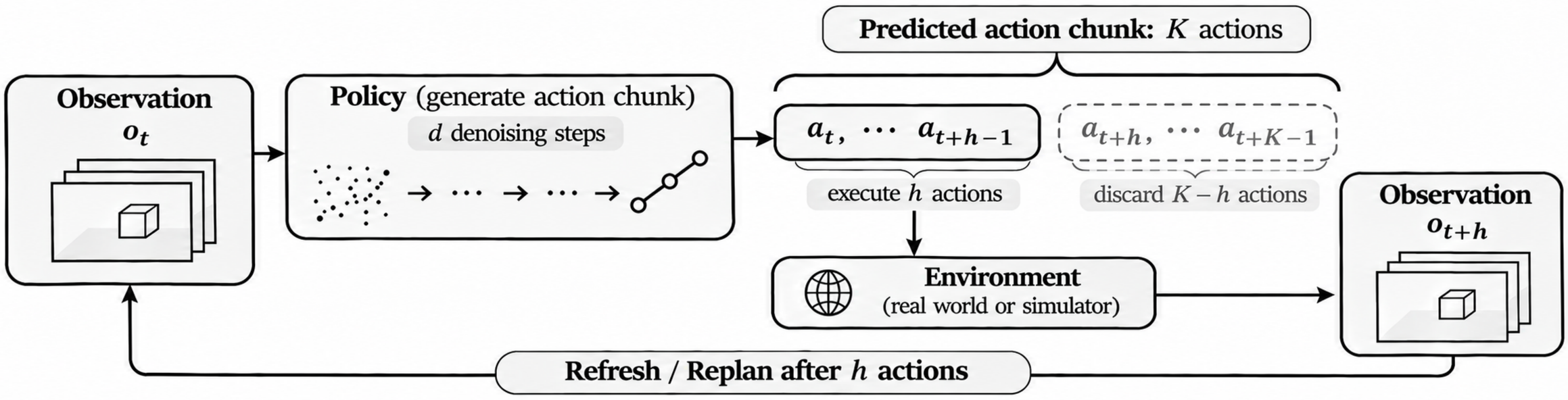
Refinement vs. Refresh

A Reproducible Evaluation Protocol for Closed-Loop Robot Policies

Alba María Téllez Fernández, MAAT Institute (alba.tellez@maatinstitute.com)

Closed-Loop Controls: Refinement and Refresh



Parameters | K : number of predicted actions per chunk (chunk length) | h : executed actions per replan ($h \leq K$) | d : denoising steps (refinement)

Synchronous closed-loop execution. Each policy call predicts K actions; only the first h are executed before obtaining a new observation and replanning.

Reproducible Protocol: Separate Design, Measurement, and Evidence Status

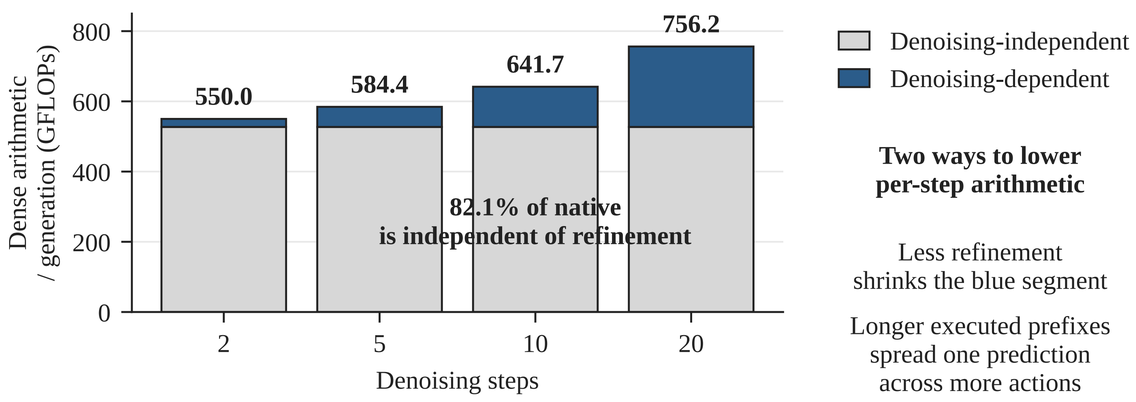
1. Define — Fix the checkpoint and all paired rollout conditions.

2. Account — Measure the full cost of each prediction and episode.

3. Freeze — Freeze primary comparisons before held-out evaluation.

4. Interpret — Label amended contrasts descriptive and limit claims.

Complete Compute Accounting



Worked Example: Frozen SmolVLA

Figure 2.1 — Held-out settings

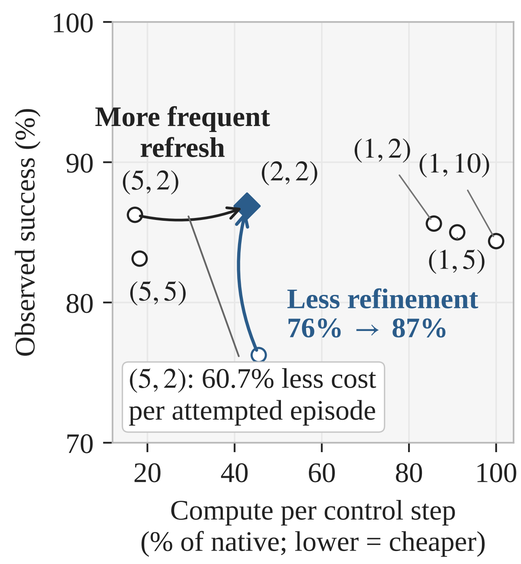
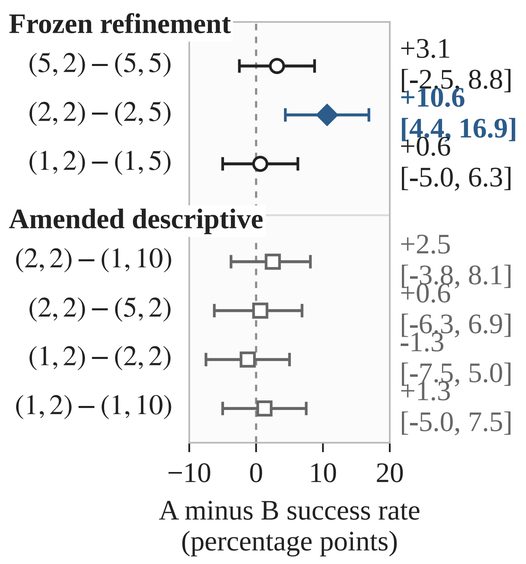


Figure 2.2 — Paired effects (A minus B)



Evidence Status: What the SmolVLA Example Supports

Frozen before confirmation

At fixed refresh frequency, reducing denoising from five to two steps increased success from 122/160 to 139/160. This is the prespecified primary comparison.

Amended: descriptive only

Refreshing after five rather than two actions produced 138/160 versus 139/160 successes with 60.7% lower arithmetic per attempt. Similar observed success does not prove equal quality, and arithmetic is not deployment latency.

SmolVLA is a worked example, not a universal allocation rule. Report success and complete computational expenditure together.
Keep confirmatory and descriptive evidence visibly separate.